

Learning Resource

1. Web Scraping (Python Requests + BeautifulSoup)

- **Real Python – Beautiful Soup: Build a Web Scraper With Python**
A detailed walkthrough for extracting data from static HTML using Requests and BeautifulSoup. Great for hands-on practice.
<https://realpython.com/beautiful-soup-web-scraper-python/>
 - **ScrapingBee – Python Web Scraping with Beautiful Soup**
Covers fundamental techniques including CSS selectors, pagination, error handling, and best practices.
<https://www.scrapingbee.com/blog/python-web-scraping-beautiful-soup/>
 - **Dataquest – Tutorial: Web Scraping with Python Using Beautiful Soup**
A beginner-oriented guide that starts with the basics and evolves into real scraping tasks.
<https://www.dataquest.io/blog/web-scraping-tutorial-python/>
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2. Database Schema Design & Planning

- **PlanetScale – Database Schema Design 101 for Relational Databases**
Introduces key concepts like tables, relationships, ER diagrams, and schema best practices.
<https://planetscale.com/blog/schema-design-101-relational-databases>
 - **Integrate.io – Complete Guide to Database Schema Design**
Explores schema types, normalization, naming conventions, and database maintenance strategies.
<https://www.integrate.io/blog/complete-guide-to-database-schema-design-guide/>
 - **Lucidchart – Database Design and Structure Tutorial**
Step-by-step coverage of entity modeling, normalization, and ER diagrams.
<https://www.lucidchart.com/pages/tutorial/database-design-and-structure>
 - **Wikipedia – Database Normalization**
Core theoretical overview of normalization forms and their importance.
https://en.wikipedia.org/wiki/Database_normalization
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3. Data Cleaning & Ingestion Concepts

- **Microsoft Support – Database Design Basics**
Practical guidance on table planning, relationships, and normalization in real-world scenarios.
<https://support.microsoft.com/en-us/office/database-design-basics-eb2159cf-1e30-401a-8084-bd4f9c9ca1f5>
 - **Wikipedia – Database Design**
Academic overview of the data modeling process, ER modeling, normalization phases, and physical schema design.
https://en.wikipedia.org/wiki/Database_design
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4. Sentiment Analysis Modeling

- **Stack Abuse – Python for NLP: Sentiment Analysis with Scikit-Learn**

Applies scikit-learn to classify sentiment from textual data, offering code examples and practical insights.

<https://stackabuse.com/python-for-nlp-sentiment-analysis-with-scikit-learn/>

- **Scikit-Learn – Working with Text Data (Official Guide)**

Covers TF-IDF, preprocessing, pipelines, and text classification fundamentals.

https://scikit-learn.org/1.4/tutorial/text_analytics/working_with_text_data.html

- **Twilio – Making Sentiment Analysis Easy With Scikit-Learn**

Demonstrates how to build a logistic regression-based sentiment classifier using tweet data.

<https://www.twilio.com/en-us/blog/sentiment-analysis-scikit-learn-html>